

FINAL REPORT



Energy Consumption & Energy Communities in AMP

E-REDES Open Data · AdEPORITO · Group C

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AGENDA

1 Business Understanding

2 Data Understanding

3 Data Preparation

4 Modelling

5 Impact Analysis

6 Conclusions

CONTEXT & END-USER NEEDS



E-REDES

Distribution System Operator (DSO)

Launched Open Data Portal in Nov 2022 to promote transparency and support the energy transition in Portugal.



AdEPORTO

Porto Energy Agency

Non-profit est. 2007, covering Porto Metropolitan Area north of Douro, over 1.1 million people.



THE BIG QUESTION

Did the implementation of energy communities have an impact on electricity consumption patterns in the AMP?

Monthly Consumption

ACC/CER Communities

DEFINING ACC/CER

**ACC***AutoConsumo Coletivo*

Collective Self-Consumption

Group of consumers sharing energy produced by a local installation (e.g. rooftop solar).

Key features:

- Members share the same building or nearby premises
- Energy is self-produced and distributed internally
- Reduces grid dependency and electricity bills

**CER***Comunidade de Energia Renovável*

Renewable Energy Community

Larger-scale community that collectively produces, consumes, stores and sells renewable energy.

Key features:

- Open to citizens, SMEs and local authorities
- Governed by community members democratically
- Enables broader energy sharing across a parish

BUSINESS OBJECTIVES & SUCCESS CRITERIA



Main Goal

Measure the real impact of ACC/CER implementation on electricity consumption in the Porto Metropolitan Area.



Analyze Consumption Patterns

Study monthly electricity consumption patterns before ACC/CER implementation across municipalities and parishes.



Prepare Predictive Dataset

Clean, join and engineer features from the consumption and CER datasets to build a robust modelling-ready dataset.



Compare Predicted vs Observed

Compare predicted post-ACC/CER consumption with observed real values to isolate the CER effect.



Evaluate ACC/CER Impact

Quantify and map the impact of energy communities at municipality and parish level for actionable insights.

REQUIREMENTS



CER Linkage to Geographic Units

ACC/CER records must link to consumption data via shared municipality and parish codes.



Sufficient Temporal Overlap

Consumption data must cover both pre-CER (2020–2022) and post-CER (2022+) periods with enough months before and after each community's start date.



Monthly Time-Series Granularity

Consumption must be available monthly to support trend modelling and seasonal analysis.



AMP Geographic Scope Filter

The analysis must be restricted to municipalities within the Porto Metropolitan Area (AMP).



ACC/CER Implementation Dates & Counts

Energy community records must include creation dates and counts per parish to define treatment periods.

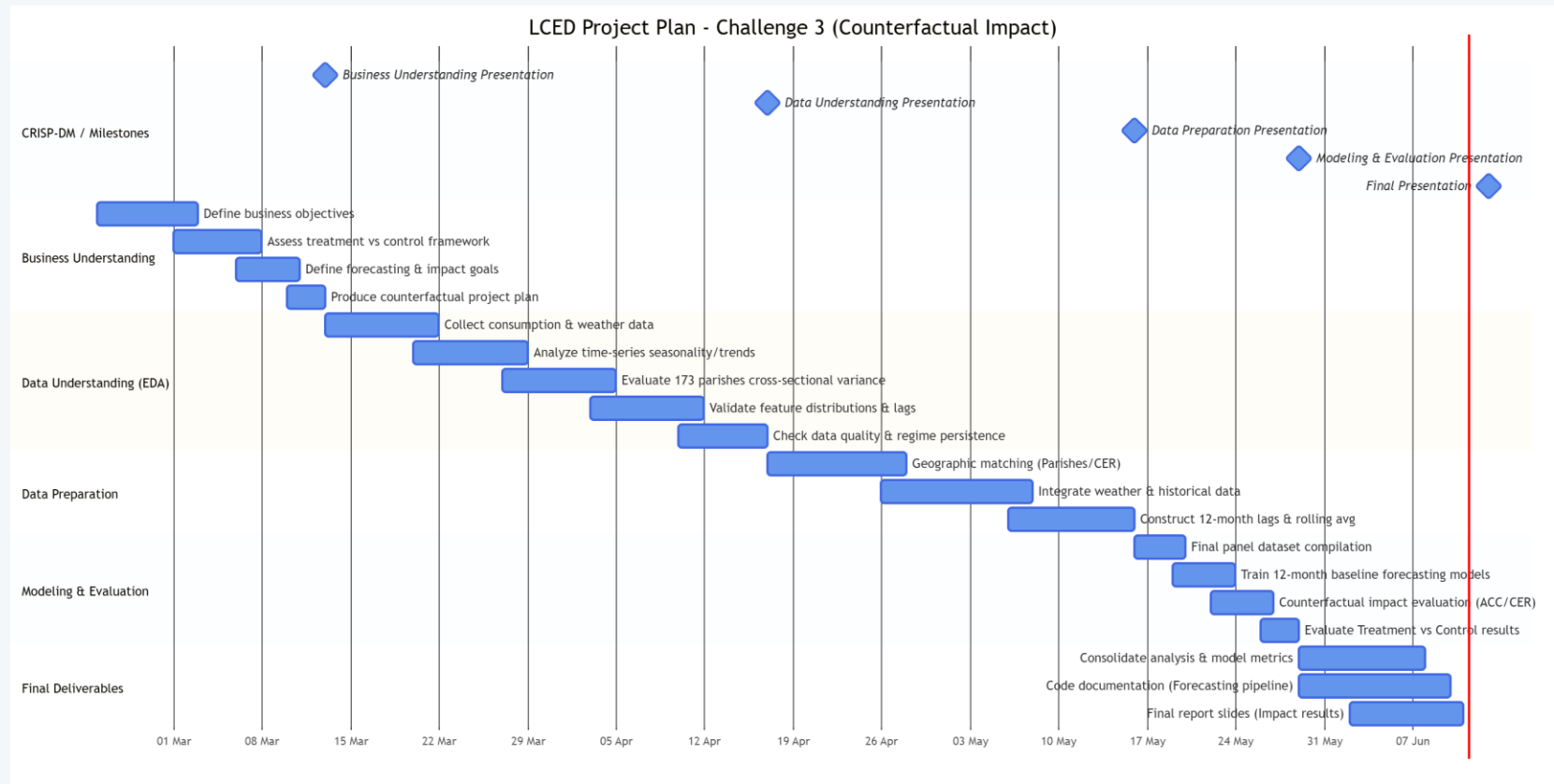


Data Quality & Consistency

No systematic gaps; missing values identifiable and geographic codes consistently encoded across both datasets.

PROJECT PLANNING

Methodology: CRISP-DM | Business Understanding → Data Understanding → Data Preparation → Modeling & Evaluation → Interpretation



Tools & Collaboration

Python Ecosystem

Pandas, NumPy, Scikit-learn, Matplotlib, Statsmodels

Jupyter Notebooks

Interactive analysis, exploratory workflows, reproducible reports

Version Control

GitHub for code history, OneDrive for shared documents

DATA SOURCES & ASSESS SITUATION: DATASET 1



Dataset 1 – Monthly Electricity Consumption by Parish

268,520

Total Records

2020-2025

Time Coverage

2 Voltage Levels

BT & MAT/AT/MT

Parish-level

Granularity

Columns: Year, Month, Date, District, Municipality, Parish, Voltage Level, Active Energy (kWh), Geographic Codes

AMP: 19,165 records across 17 municipalities, from Nov 2020 onwards.

	Year	Month	Date	District	Municipality	parish	Voltage level	Energia Ativa	coddistrito	coddistritoconcelho	DistrictMunicipalityParishCode	mes_int	in_treatment
2	2022	2	2022-02-01	Porto	Valongo	VALONGO	Muito Alta, Alta e Média Tensões	1330171.291	13	1315	131505	2	True
4	2022	8	2022-08-01	Aveiro	Santa Maria da Feira	UF CALDAS SAO JORGE PIGEIOS	Muito Alta, Alta e Média Tensões	288588.601	1	0109	010932	8	True
45	2021	8	2021-08-01	Porto	Gondomar	UF FANZERES SAO PEDRO COVA	Muito Alta, Alta e Média Tensões	1578134.211	13	1304	130413	8	True
46	2025	3	2025-03-01	Porto	Gondomar	UF MELRES E MEDAS	Muito Alta, Alta e Média Tensões	1798638.306	13	1304	130416	3	False
63	2023	8	2023-08-01	Porto	Paredes	PARADA DE TODEIA	Muito Alta, Alta e Média Tensões	193248.617	13	1310	131017	8	False

DATA SOURCES & ASSESS SITUATION: DATASET 2



Dataset 2 – Energy Communities (ACC/CER) Implementation Records

6,636

Total Records

2022-2025

Time Coverage

ACC & CER

Community Types

Parish-level

Granularity

Columns: Date, Year, Month, District, Municipality, Parish, ACC/CER Type, Count, Geographic Codes

AMP: 1,060 records (ACC: 1,546 | CER: 25) across 16 municipalities, from 2023 onwards.

	Date	Year	Month	District	Municipality	Parish	Type ACC/CER	ACC_CER_Count	District Code	Municipality Code	Parish Code
5	2025-09-01	2025	9	Porto	Maia	Cidade da Maia	ACC	2	13	1306	130619
7	2025-09-01	2025	9	Porto	Valongo	Ermesinde	ACC	4	13	1315	131503
14	2025-09-01	2025	9	Porto	Vila do Conde	Guilhabreu	ACC	1	13	1316	131611
18	2025-09-01	2025	9	Porto	Vila do Conde	Macieira da Maia	ACC	1	13	1316	131614
20	2025-09-01	2025	9	Aveiro	Santa Maria da Feira	Milheirós de Poiares	ACC	1	1	109	010914

DATA SOURCES & ASSESS SITUATION: EXTERNAL DATA



External Data - Weather

87,648

Total Records

2021-2025

Time Coverage

t2m & tp

Variables

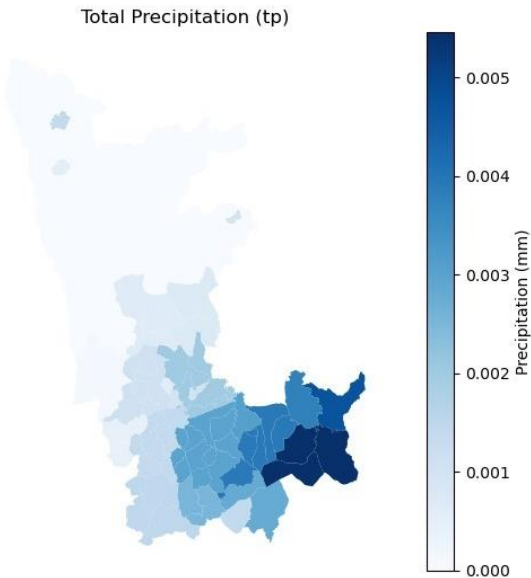
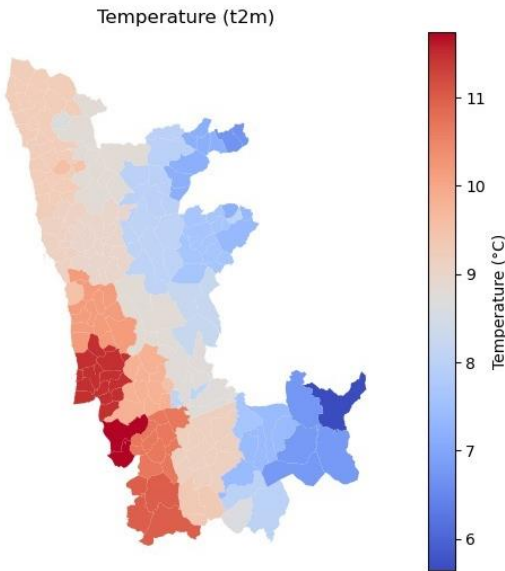
Hourly

Resolution

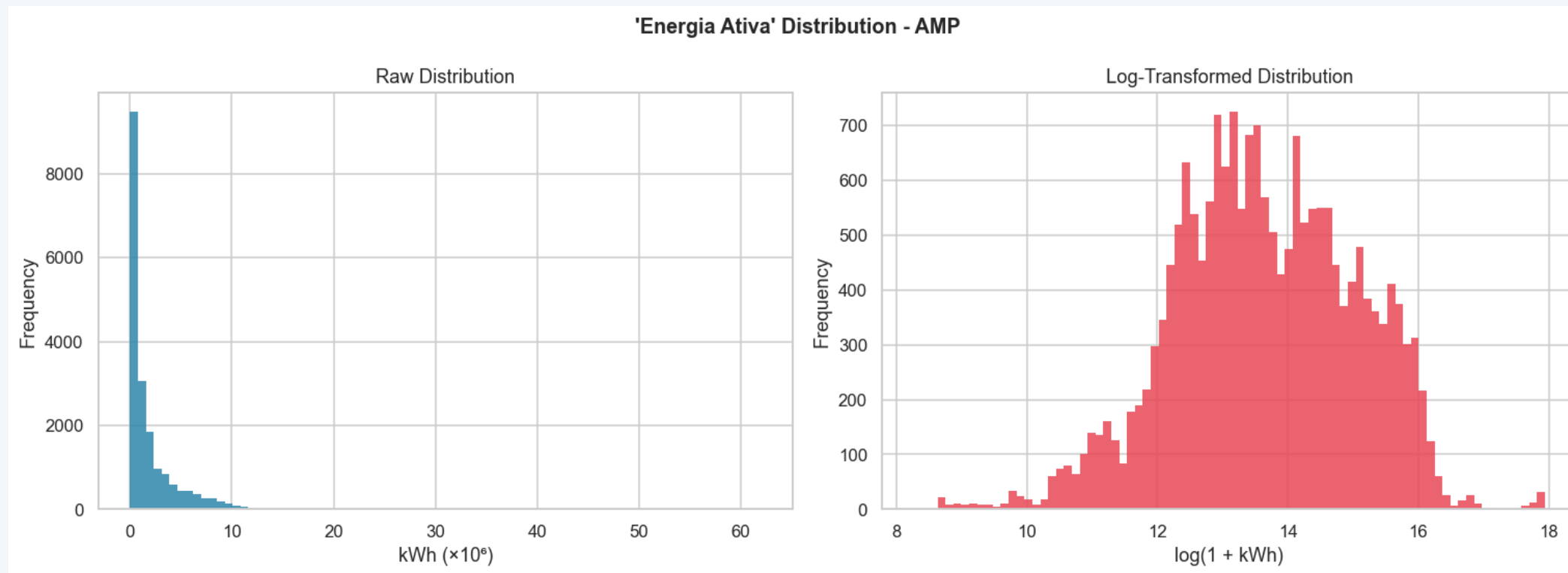
Format: NetCDF (.nc) • Variables: t2m (2m temperature), tp (total precipitation)

Source: Copernicus Climate Data Store (ERA5). 43,824 hourly records per variable, from 2021 to 2025

Example of temperature (left) and total precipitation (right) values in AMP (1st of January 2025 at 6 PM)



MONTHLY ELECTRICITY CONSUMPTION: DISTRIBUTION



Highly Skewed Raw Data

The raw electricity consumption data is heavily right-skewed (skewness of 8.39), indicating that most parishes have lower consumption levels, while a few extreme outliers consume massive amounts.

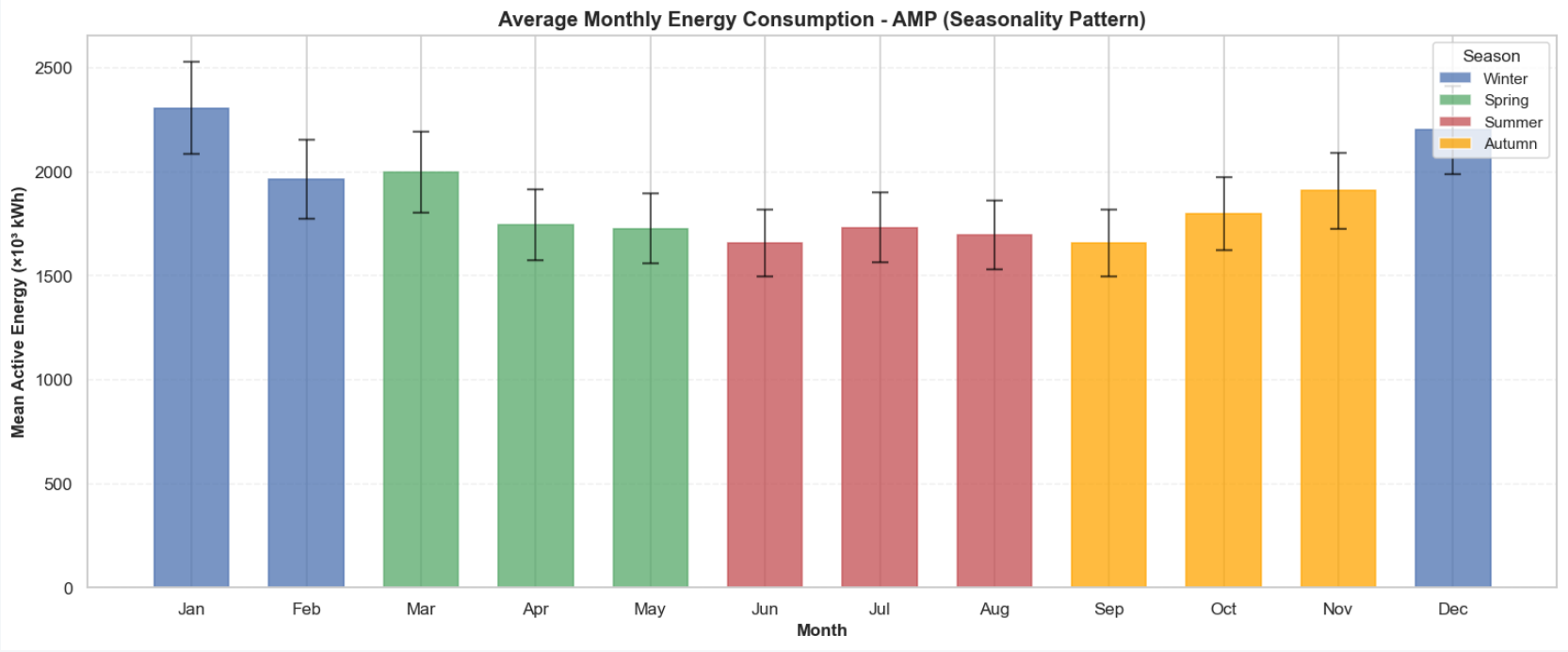
Mean vs. Median Disparity

The mean consumption is 2,037,021 kWh, which is substantially higher than the median of 795,647 kWh. This large gap, along with a maximum value of 62,000,970 kWh, highlights the strong pull of high-consumption outliers.

Effective Log Transformation

Applying a log transformation successfully normalizes the data, dropping the skewness to nearly zero (-0.18). This transformation results in a much more symmetric, bell-shaped distribution that is better suited for standard statistical modeling.

MONTHLY ELECTRICITY CONSUMPTION: SEASONALITY



Key Findings

Winter Peak vs. Summer Trough

January records the highest average consumption (2,324,764 kWh), while August has the lowest (1,805,264 kWh).

Data Stability

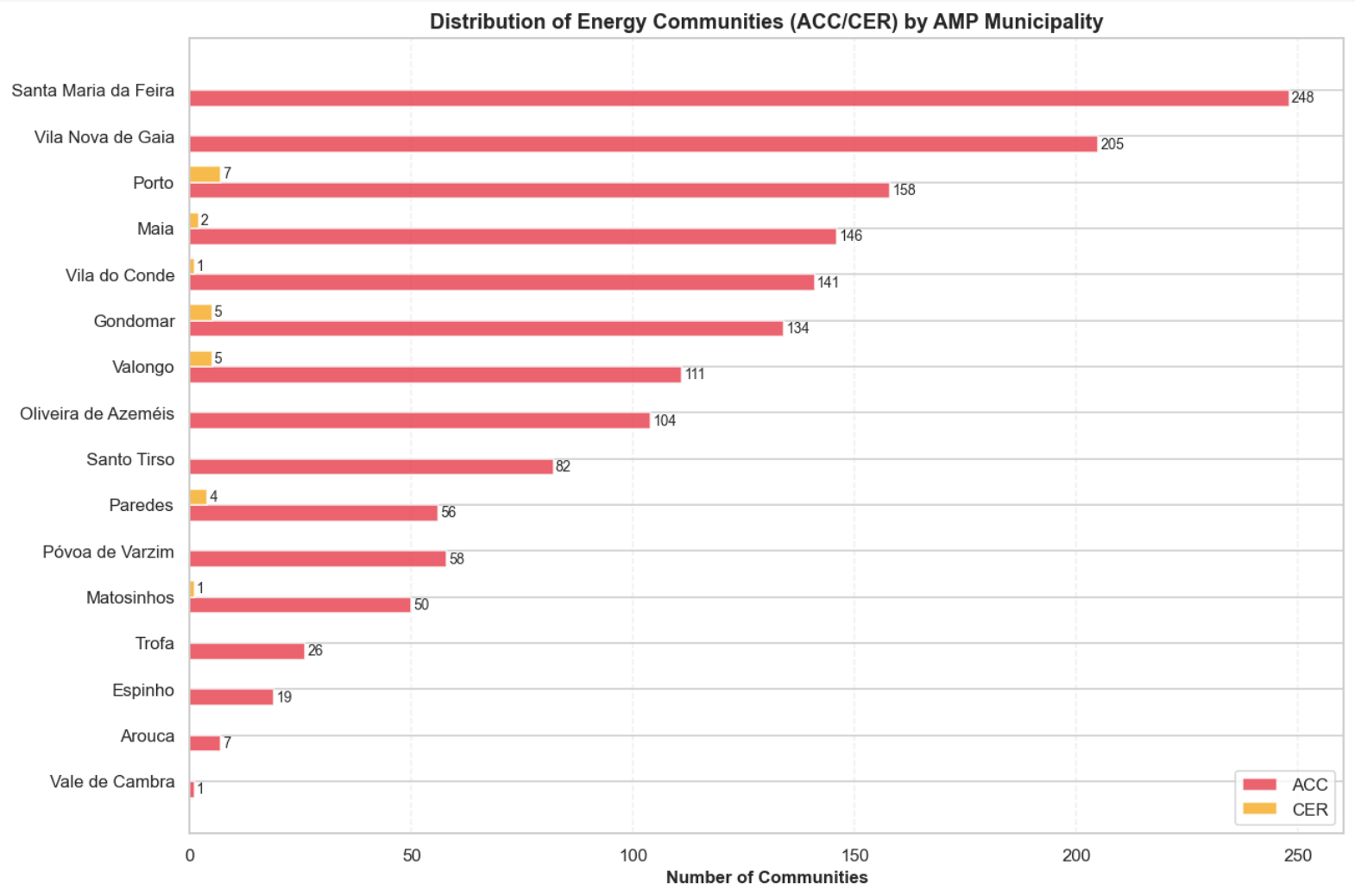
The coefficient of variation remains relatively stable across all months, pointing to a consistent behavioral pattern over time.

Statistical Validation

Confirmed Seasonality

Kruskal-Wallis tests prove that consumption differences across both months and seasons are highly statistically significant (p-value < 0.001).

ENERGY COMMUNITIES (ACC/CER): DISTRIBUTION



Key Findings

ACC Dominance

The landscape is almost entirely driven by ACC. Across the entire AMP, there are 1,546 ACCs compared to a mere 25 CERs.

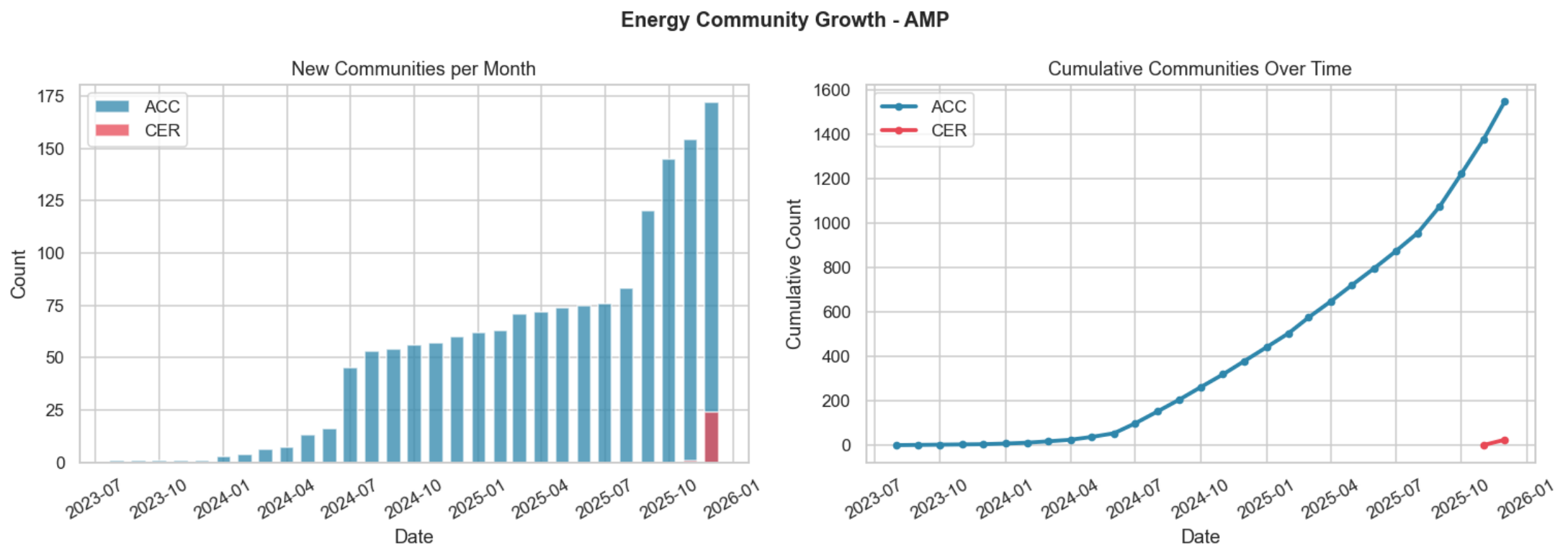
Geographic Concentration

Adoption is highly concentrated in a few key areas. Santa Maria da Feira (248) and Vila Nova de Gaia (205) lead the region by a significant margin, exclusively with ACCs.

Incipient CER Adoption

CERs are still very rare and are only present in 7 out of the 17 municipalities. Porto leads this specific segment with 7 CERs, followed by Gondomar (5) and Valongo (5).

ENERGY COMMUNITIES (ACC/CER): GROWTH OVER TIME



Exponential ACC Adoption

The cumulative chart demonstrates a clear exponential growth trend for ACCs. Adoption starts slowly in late 2023, begins accelerating in mid-2024, and reaches a total of 1,546 communities by the end of the observed period (December 2025).

Late 2025 Surge

The monthly breakdown reveals a massive spike in new community registrations during the final quarter of 2025, indicating a rapidly accelerating market or a backlog of approvals being cleared.

Late CER Emergence

CERs are entirely absent for almost the entire timeline. They only begin to appear at the very end of 2025, culminating in just 25 active communities.

DATA QUALITY ASSESSMENT

1. Data Completeness

- **Ideal Panel Coverage:** The panel shows 99.9% completeness. The strong pre-treatment history (starting Nov 2020) provides a reliable baseline and supports lagged variables. The temporal mismatch at the end of 2025 is a non-issue, as the causal impact evaluation naturally focuses on pioneer communities (2023/2024) with sufficient pre/post-treatment data.
- **Spatial Coverage:** All 17 AMP municipalities are represented, enabling the creation of a robust, parish-level control group.

2. Data Correctness & Errors

- **Negative Energy Values:** Physically impossible consumption values (≤ 0), traced back to OUTROS mapping failures.
 - **Solution:** Addressed by excluding OUTROS records.
- **Extreme Outliers:** The Active Energy (*Energia Ativa*) distribution is heavily right-skewed.
 - **Solution:** Apply a logarithmic transformation to normalize the data and stabilize the variance of the time series.

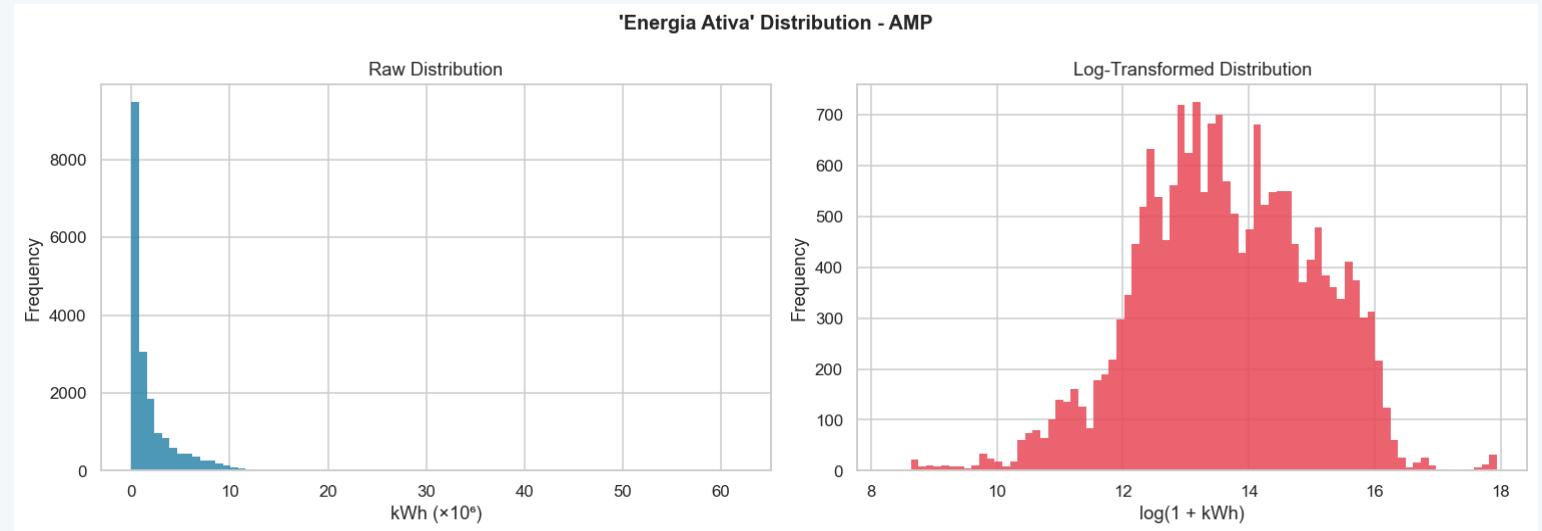
3. Missing & Unmapped Values

- **No Standard Nulls:** Both datasets are 100% complete regarding standard missing values (NaNs).
- **Mapping Failures ("OUTROS"):** Records without municipality/parish codes, making them both spatially and physically invalid.
 - **Solution:** Exclude all OUTROS records

CONSUMPTION & ENERGY COMMUNITIES DATA

Cleaning & Transformations

- **Unit check:** Use kWh for the energy
- **Parish names:** Standardize parish names
- **Quality Control:** Filtered out unmapped "OUTROS" registry failures to eliminate negative consumption anomalies
- **Filtering:** Filter only for the parishes in Porto Metropolitan Area
- **Energy Transformation:** Energy consumption was heavily right-skewed, so a $\log(1+x)$ transformation was applied to normalize the distribution



Join

Consumption + Energy Communities

- **Primary Key:** 6-digit Parish Code (DICO).
- **Temporal Key:** Year-Month string.

Result: Parishes with active communities received a flag (`'is_active=1'`), while non-community parishes defaulted to control (`'0'`).

Important new columns

- ***acc_cer_start_date*:** Data of 1 active energy community in parish
- ***has_acc_cer*:** 0 if no energy community, 1 if treated already
- ***months_since_acc_cer*:** Months until (negative) or since (positive) the acc/cer
- ***control*:** True if control parish, false if treated parish
- ***n_acc_cer*:** Current number of active energy communities

ENVIRONMENTAL DATA JOIN

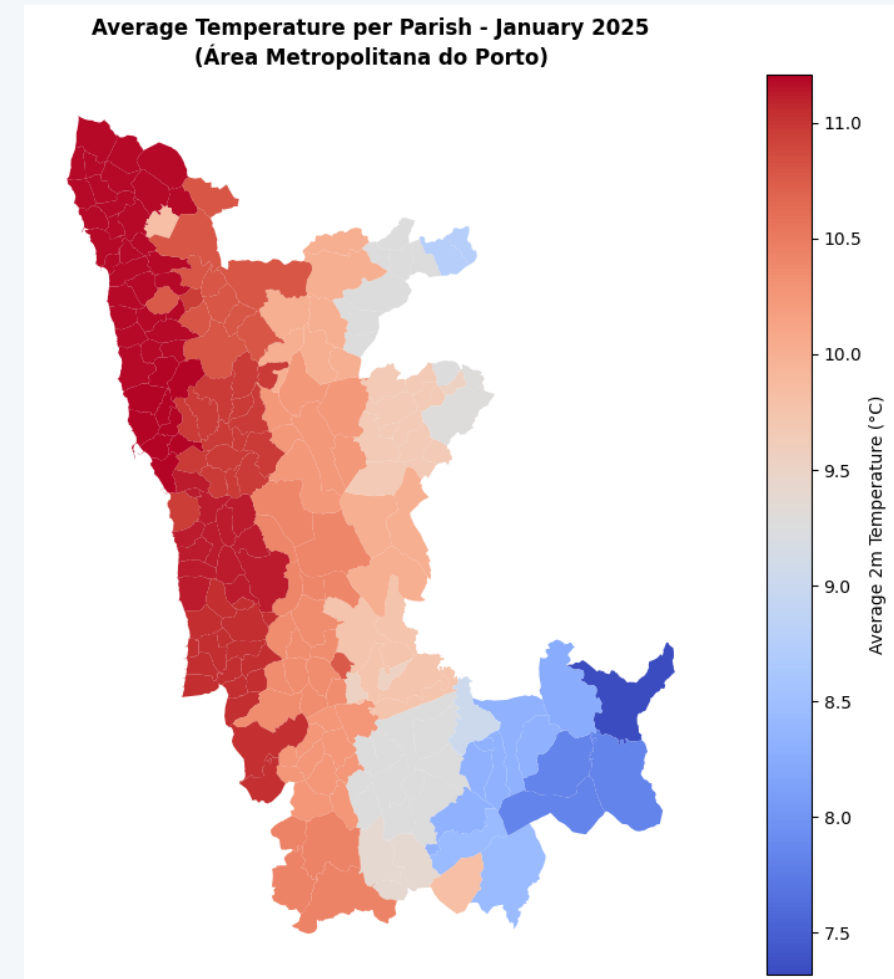


Temperature and Precipitation

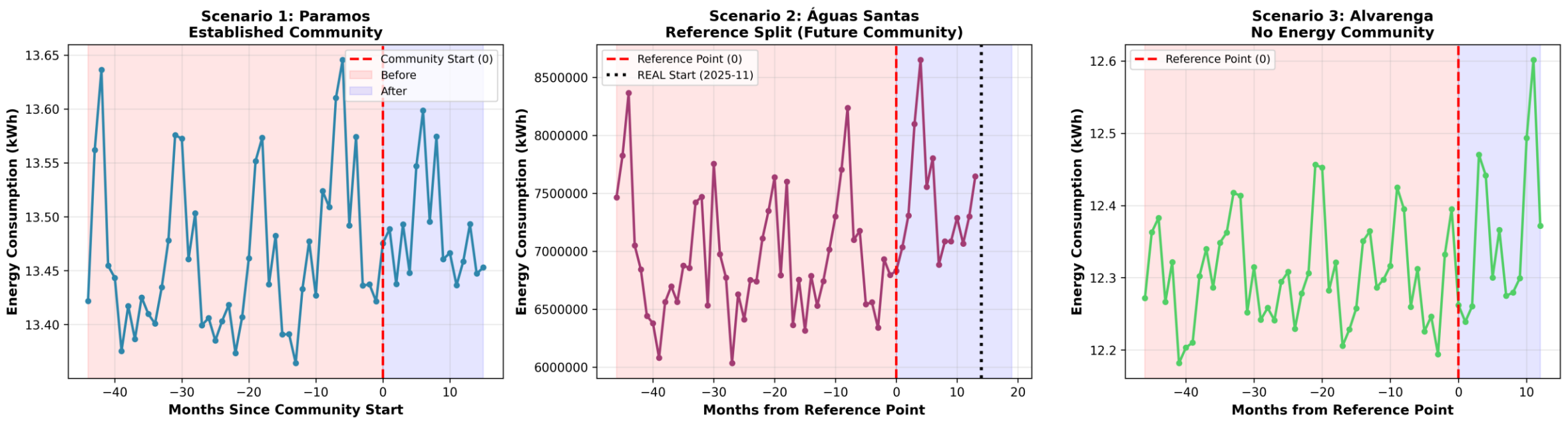
- Performed a Spatial Join mapping each parish's geographic polygon to the nearest ERA5 coordinate grid point before aggregating the hourly data into monthly values
- Join into consumption & energy communities using parish codes

Variables:

- 2m temperature (°C) : Monthly average per parish
- Total precipitation (mm) : Accumulated monthly values per parish



PARISH SCENARIOS



Scenario 1

Established: Established community with clear Pre and Post intervention data available (at least 12 months post).

Scenario 2

Future: Community starting after our consumption data window (energy community not used), and the consumption data used as in scenario 3.

Scenario 3 (Control)

Never: Parishes with no energy communities serve as our baseline. Defined a reference point (t=0) where there are 12 months of energy data.

PRE & POST INTERVENTION DATASETS

Pre Active Community

Covers the timeline strictly before any energy community was registered in a given parish.

7,851 Records

Purpose: Used by the models to learn the standard consumption patterns, seasonal trends, and baseline behavior without community intervention.

Post Active Community

Covers the timeline starting from the exact month the first energy community became active in that parish.

2,420 Records

Purpose: Used to evaluate counterfactuals - comparing actual consumption against what the models predict would have happened without the ACC/CER.

FEATURE ENGINEERING



Lags & Rolling Averages

Smoothing out noise and capturing historical baselines

- **lag_12:** Exact consumption 12 months prior
- **roll_3 / 6 / 12 / 24:** Moving averages over multiple windows for short and long-term trends



Momentum & Growth

Quantifying the rate of change in energy usage over time

- **momentum:** Short-term directional shifts in consumption
- **yoy_ratio:** Year-over-year relative growth rate



Calendar & Variance

Accounting for external temporal factors and instability

- **volatility:** Rolling variance to measure consumption stability
- **holiday_rate:** Proportion of holidays in a given month affecting baseline usage
- **cyclical encoding:** Created month_sin and month_cos variables to cyclically encode the months

MODELLING: PROBLEM FORMULATION

Supervised regression: one prediction per parish

- For each parish we predict the average electricity consumption over a 12-month window, using only data strictly before that window.
- The trained model then plays a counterfactual role for the impact analysis (what the consumption would have been without an energy community).

X

Features: independent variables

The last ~13 months of each parish's history before the cutoff: lag_12, roll_3/6/12/24, momentum, yoy_ratio, slope, volatility, weather means, holiday rate, calendar.

$\hat{Y}$

Model Prediction: expected mean for the 12 months after creation

The trained model's forecast of what the next 12 months would look like without intervention. This is the counterfactual baseline used downstream in the impact analysis.

Y

Training Target: mean of the 12 months before community creation

For each parish, the average kWh over the 12 months ending at its cutoff date (= the month before CER appears, or end-of-data for control parishes).

Y_real

Observed Reality (only used at the impact-analysis stage)

The actual consumption recorded after the CER community was set up.
Impact estimate = $Y_{\text{real}} - \hat{Y}$ (with bias correction from controls).

MODELLING: PROBLEM FORMULATION

X

Features: independent variables

Y

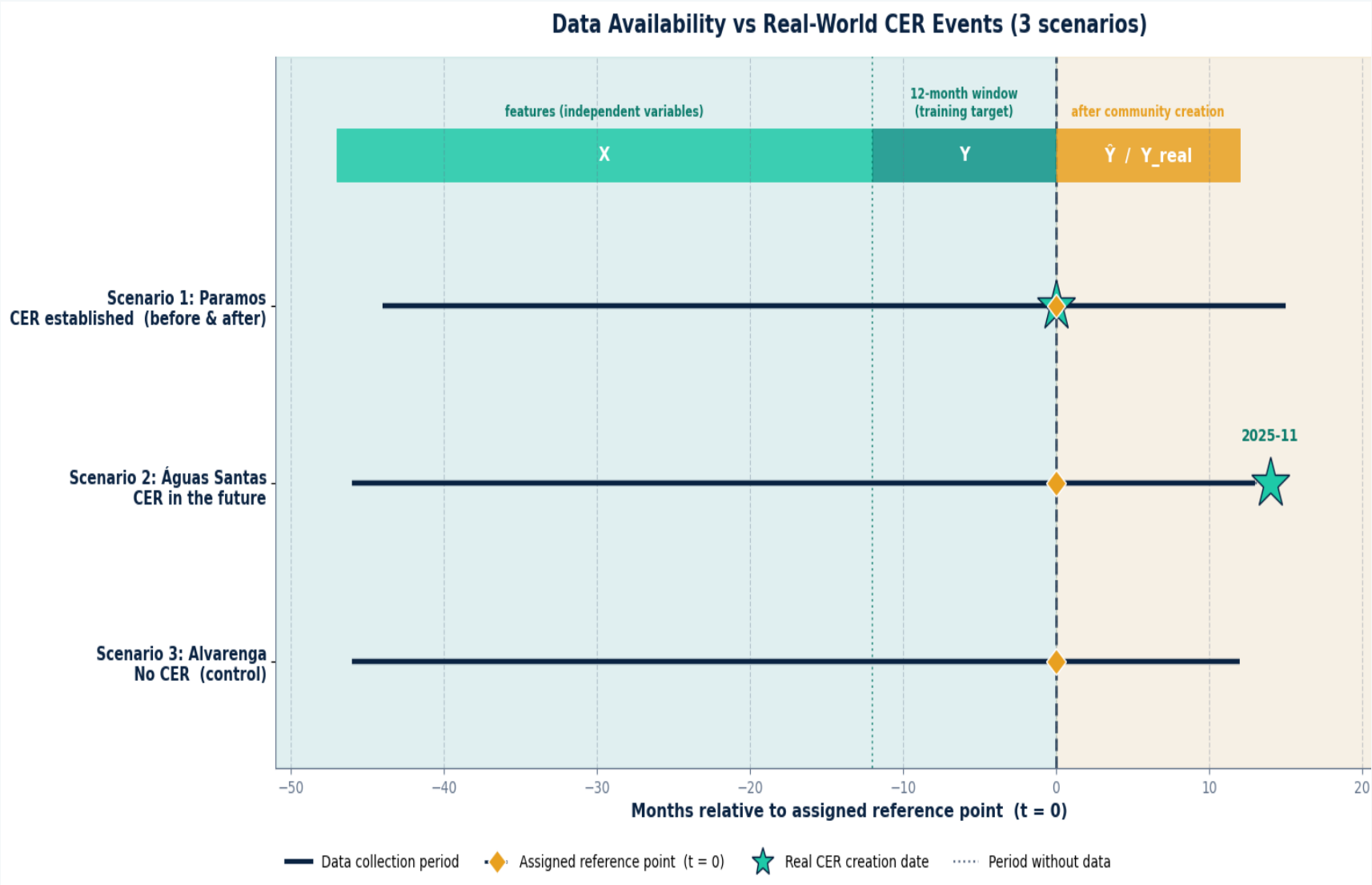
Training Target: mean of the 12 months before community creation

$\hat{Y}$

Model Prediction: expected mean for the 12 months after creation

Y_{real}

Observed Reality (only used at the impact-analysis stage)



Three real parishes. The X / Y / $\hat{Y}$ / Y_{real} bands at the top show where each variable lives on the timeline.

TEMPORAL SPLIT & FEATURE ENGINEERING

Leakage rule

No information from the target window may appear in any feature: all preprocessing fit only on training partition.

PER-PARISH TEMPORAL SPLIT

- Each parish has its own cutoff date = first day of its target window
- Training set = months strictly before cutoff
- Asserted programmatically:
 $\max(\text{input_date}) < \min(\text{target_date})$ for all parishes

CROSS-SECTIONAL FEATURES (one row per parish)

Lag / level: lag_12, lag_24, roll_3, roll_6, roll_12, roll_24

Momentum: momentum_3_6, momentum_6_12, yoy_ratio, slope

Volatility: std, coefficient of variation

Weather: recent_t2m, recent_tp, hist_t2m, hist_tp, t2m_trend

Calendar: holiday_rate, last_month

173

Parishes

~30 - 48

Train months / parish

21

Cross-sectional features

12

Target months / parish

TWO-LAYER LEAKAGE PROTECTION

LAYER 1

Within each parish

No future month of the same parish leaks into its own features.

LAYER 2

Across parishes

Older-CER parishes train, newer-CER parishes validate (see next slide).

CROSS-VALIDATION DESIGN: STRICT FORWARD CHRONOLOGICAL

Train only on parishes whose ACC/CER appeared earlier than the validation parishes.

WHY RANDOM K-FOLD LEAKS

Random folds mix early-CER and late-CER parishes in the same training set, so the model can use information from parishes "in the future" to predict parishes in the past.

This inflates ML scores and lets ensembles look stronger than they really are on out-of-sample data.

CHRONOFORWARDSPLIT (CUSTOM CV)

Parishes are sorted by CER appearance and split into chronological bins. Training set always contains parishes whose CER was **earlier** than the parishes being validated.

Strict forward order, no future leakage.

NESTED CV PRESERVED

Hyperparameters are tuned *inside* each outer training set, never on the validation parishes, so the leaderboard score is honest.

86

Warm-up parishes

5 folds

Validation

17 - 18

Parishes per fold

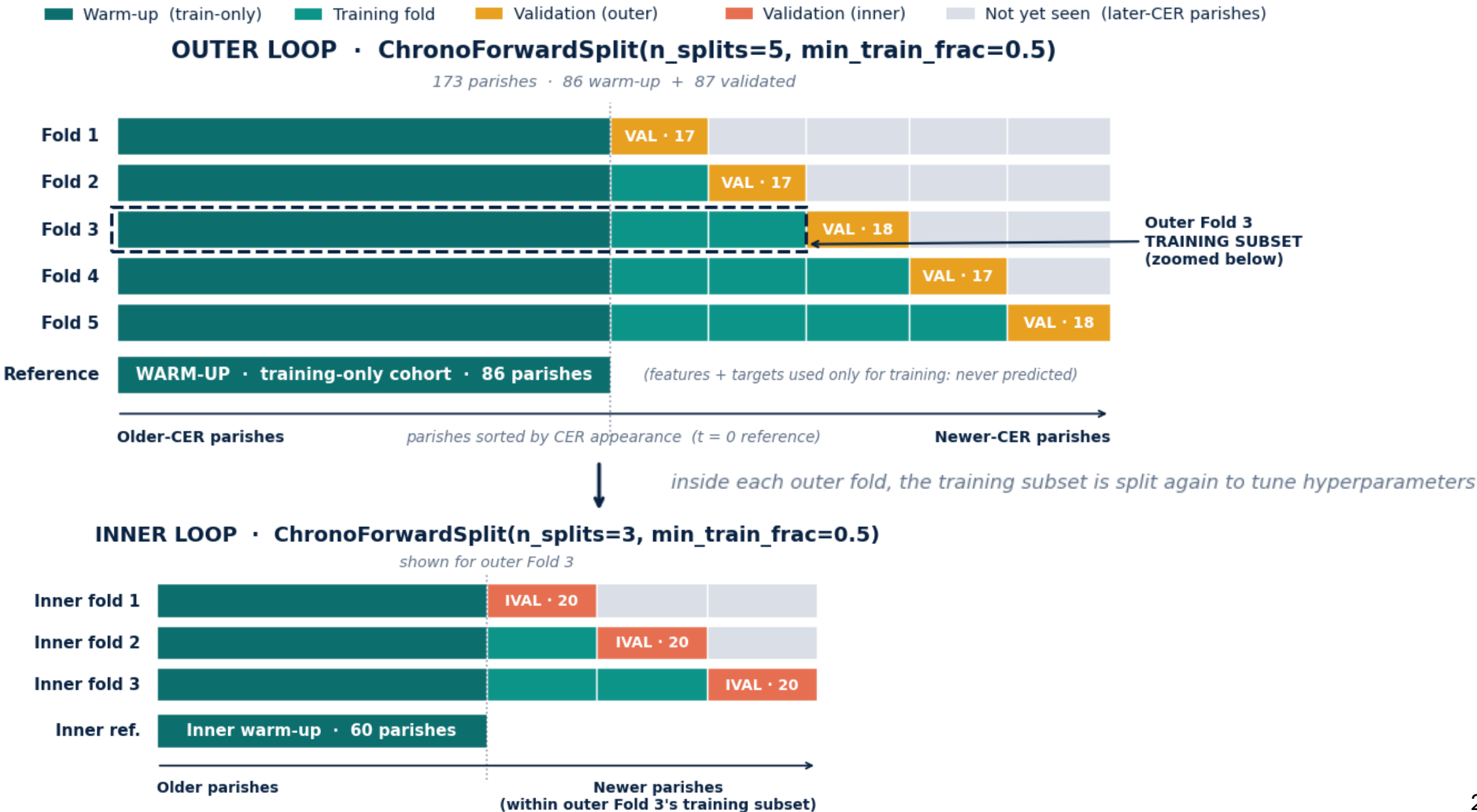
87 / 173

Parishes scored
in the leaderboard

Honest cross-validation

Coverage trade-off: ~50% of parishes are training-only warm-up; the leaderboard MAE is over the remaining 87 validated parishes.

CROSS-VALIDATION DESIGN: STRICT FORWARD CHRONOLOGICAL CV



EVALUATION METRICS

Six metrics computed per model, each captures a different aspect of forecast quality.

METRIC	WHAT IT MEASURES	ROLE	PRIORITY
MAE (kWh)	Average absolute error: directly interpretable in kWh, robust to outliers	Primary error metric	✓ PRIMARY
MAPE %	Scale-free % error: comparing parishes of different sizes	Cross-parish comparison	✓ SECONDARY
MdAPE %	Median % error: robust counterpart to MAPE	Typical parish experience	✓ SECONDARY
R ²	Variance explained: how tightly the model tracks consumption patterns	Fit quality	✓ DIAGNOSTIC
Bias	Mean signed error: detects systematic over or under-prediction that MAE alone hides	Directional check	✓ DIAGNOSTIC
RMSE	Root mean squared error: penalises large errors, sensitive to outlier parishes	Outlier sensitivity	✓ DIAGNOSTIC

Primary: MAE · Comparison: MAPE / MdAPE · Diagnostic: R² and Bias · RMSE reported for completeness

NAIVE BASELINES: DETERMINING THE BENCHMARK

All four parameter-free predictors run together; lowest MAE is promoted to "the benchmark to beat".

GLOBAL MEAN

MAE 3,088,714

MAPE 241%

Same number for every parish (training-set mean).
No personalisation, no time information.

PARISH OVERALL MEAN

MAE 150,435

MAPE 4.91%

Each parish predicted as its overall historical mean.
No recency or seasonal information.

LAST VALUE

MAE 399,341

MAPE 9.30%

Each parish predicted as its most recent training observation. Highly noisy.

SEASONAL NAIVE (winner)

MAE 132,563

MAPE 4.46%

Each parish predicted as its last-12-month mean.
Captures recency AND annual seasonality.

Seasonal Naive adopted as PRIMARY BENCHMARK: every fitted model must beat its 132,563 kWh MAE.

ML CHALLENGERS: MODEL FAMILIES

The target is one scalar per parish (12-month mean). Cross-sectional is the natural setup; panel is kept as a sensitivity check.

PRIMARY SETUP

Cross-sectional · 10 models · X with 21 features

CLASSICAL TIME-SERIES

ETS (Exponential Smoothing)

LINEAR / REGULARISED

Linear Regression · Ridge
Lasso · ElasticNet

TREE-BASED (cross-sectional)

Random Forest · XGBoost
(panel variants → next column)

KERNEL / NEIGHBOUR

KNN regressor (distance-weighted)
SVR (RBF kernel)

All 10 primary models share the same X matrix, all evaluated under nested ChronoForwardSplit (5 outer × 3 inner).

SENSITIVITY CHECK

Panel formulation

one row per (parish × month)

MODELS

- LightGBM Panel
- CatBoost Panel

PURPOSE

Test whether the richer monthly structure adds signal beyond the cross-sectional aggregates.

In our run: it did not, panel does not outperform the best primary model.

Tuning protocol (shared): outer CV = ChronoForwardSplit(5, min_train_frac=0.5) · inner CV = ChronoForwardSplit(3, 0.5) · Grid/RandomizedSearch

ENSEMBLE METHODS: COMBINING SEVERAL MODELS

Two strategies for combining the per-model OOF predictions into a stronger blend.

Stacking Meta-Learner

two-layer learning

Idea: train several base models, then train a NEW model whose inputs are their predictions and whose output is the final answer.

Base models: Seasonal Naive, Parish Mean, Linear Regression, Ridge, XGBoost, Random Forest, LightGBM Panel.

Meta-learner: Ridge (α tuned by inner CV) over the OOF predictions; learns to weight the bases: "when XGBoost says X and Ridge says Y, the truth is closer to Z."

Optimised Weighted Ensemble

optimal blend

Idea: a weighted average of every model's predictions;
$$\text{final} = w_1 \cdot \text{Ridge} + w_2 \cdot \text{XGBoost} + w_3 \cdot \text{LightGBM} + \dots$$
with weights summing to 1 and ≥ 0 .

How: instead of guessing the weights, `scipy.optimize.minimize` finds the weights that minimise prediction error on held-out folds.

Result: each model contributes only as much as it actually helps; weak models get a near-zero weight automatically.

LEADERBOARD: STRICT FORWARD CHRONOLOGICAL CV

Common-subset evaluation: every model scored on the same 87 parishes; the 86 oldest-CER parishes are always in the training window under strict forward CV and never reach the validation stage.

Models trained on log-transformed kWh; metrics back-transformed to original kWh for reporting.

Model	MAE (kWh)	MAPE %	MdAPE %	R ²	Group
Optimised Ensemble ◀ winner <i>SLSQP blend: Linear Reg 63% + Parish Mean 37%</i>	67,589	3.54	2.47	0.9984	Ensemble
Linear Regression	81,950	3.96	2.35	0.9978	ML challenger
Ridge	87,068	4.11	2.89	0.9978	ML challenger
Simple Naive (parish mean)	90,259	4.90	3.59	0.9976	Sanity baseline
Lasso	101,671	4.84	4.20	0.9977	ML challenger
Seasonal Naive ◀ benchmark	103,469	5.16	3.83	0.9967	Primary benchmark
Stacking	104,621	4.54	2.92	0.9965	Ensemble
ElasticNet	106,328	4.77	3.48	0.9969	ML challenger
Random Forest	117,896	10.69	4.29	0.9953	ML challenger
SVR (RBF + log1p)	125,786	6.21	3.87	0.9958	ML challenger
ETS	128,985	5.74	5.09	0.9956	ML challenger
CatBoost Panel (sensitivity)	130,059	9.10	5.16	0.9949	Sensitivity check
LightGBM Panel (sensitivity)	203,621	13.61	7.04	0.9855	Sensitivity check
Simple Naive (last value)	210,939	9.79	8.80	0.9898	Sanity baseline
XGBoost	224,296	12.79	6.49	0.9762	ML challenger
KNN	677,314	30.91	19.28	0.8298	ML challenger

Optimised Ensemble wins (MAE 67,589 kWh): –35% vs Seasonal Naive, –25% vs Parish Mean
 But why did SLSQP collapse 9 of 11 candidates to zero? Three diagnostics in the next slide confirm this is correct.

ENSEMBLE (WINNER): WEIGHT VALIDATION & DIAGNOSTIC CHECKS

SLSQP collapsed 9 of 11 candidates to near-zero weight. Is this a principled result or a numerical accident? Three independent tests answer this.

SLSQP-OPTIMAL WEIGHTS (sum to 1.0, non-negative)

Linear Regression · 62.7 %

Parish Mean · 37.3 %

All other 9 candidates (Seasonal Naive, ETS, Ridge, Lasso, ElasticNet, XGBoost, RF, LightGBM Panel, CatBoost Panel) received weight ≈ 0 .

CHECK 1: OOF CORRELATION MATRIX

What we tested: do the 9 collapsed models correlate with Linear Reg on OOF rows?

Finding: tree models and regularised variants all correlate >0.95 with Linear Reg. Sparsity is expected: multicollinearity means those models add no new information.

CHECK 2: GREEDY FORWARD SELECTION

What we tested: build the ensemble one model at a time, adding whichever gives the biggest MAE gain.

Finding: selection naturally stops at 2 models (Linear Reg + Parish Mean). SLSQP is independently confirmed: a completely different algorithm arrived at the same answer.

CHECK 3: REGULARISED BLEND (L2 PENALTY)

What we tested: force weight spread across all candidates by adding an L2 penalty to the objective.

Finding: MAE jumps. The sparse solution is genuinely optimal, not fragile: spreading weights hurts performance, confirming the 2-model blend isn't an artefact.

All three diagnostics confirm: the 2-model blend is correct and optimal, not a numerical artefact.

ENSEMBLE (WINNER): SENSITIVITY TO LOSS FUNCTION

Does the MAE-optimal blend hold up when we re-run SLSQP under different objectives?

SLSQP-OPTIMAL WEIGHTS (sum to 1.0, non-negative)

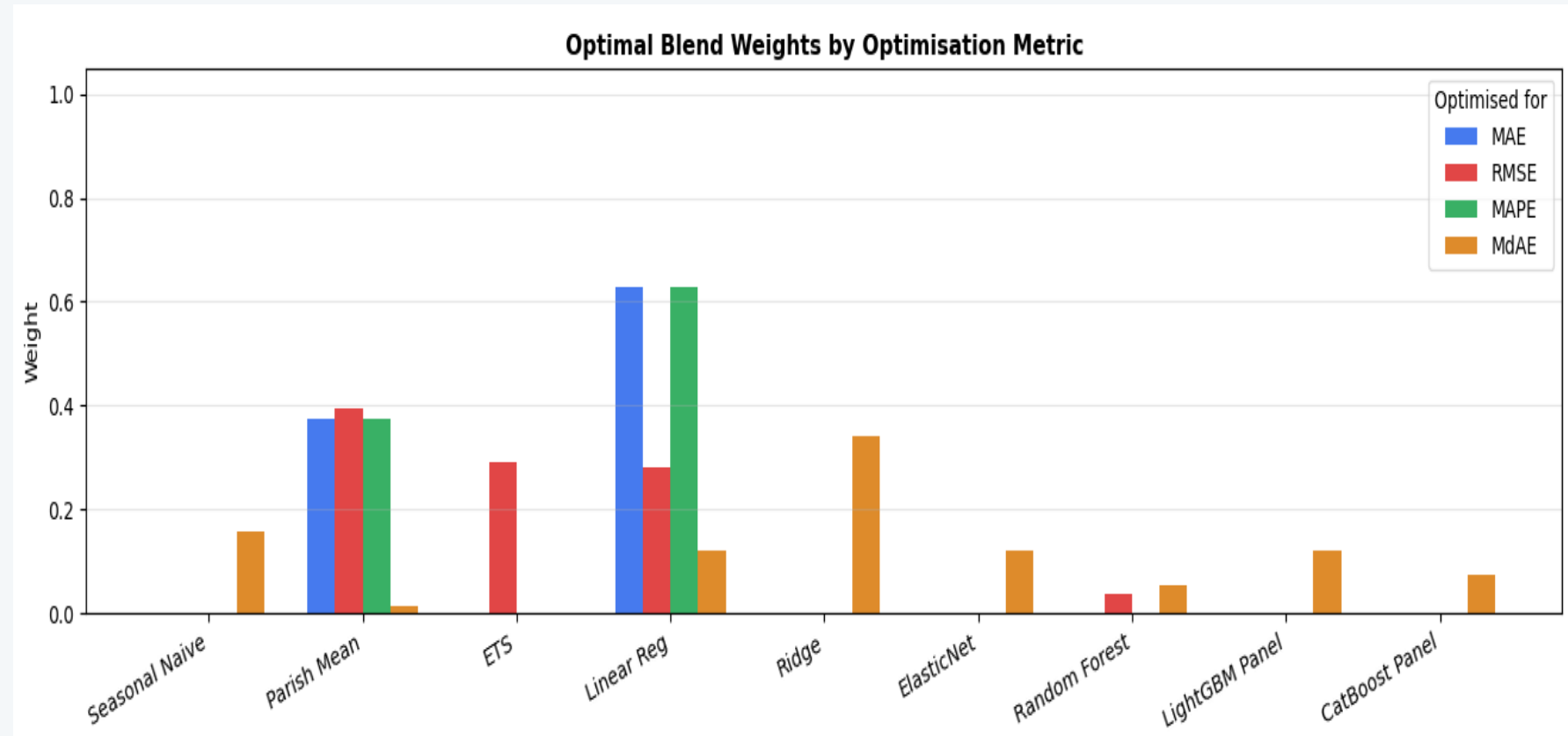
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Parish Mean · 37.3 %

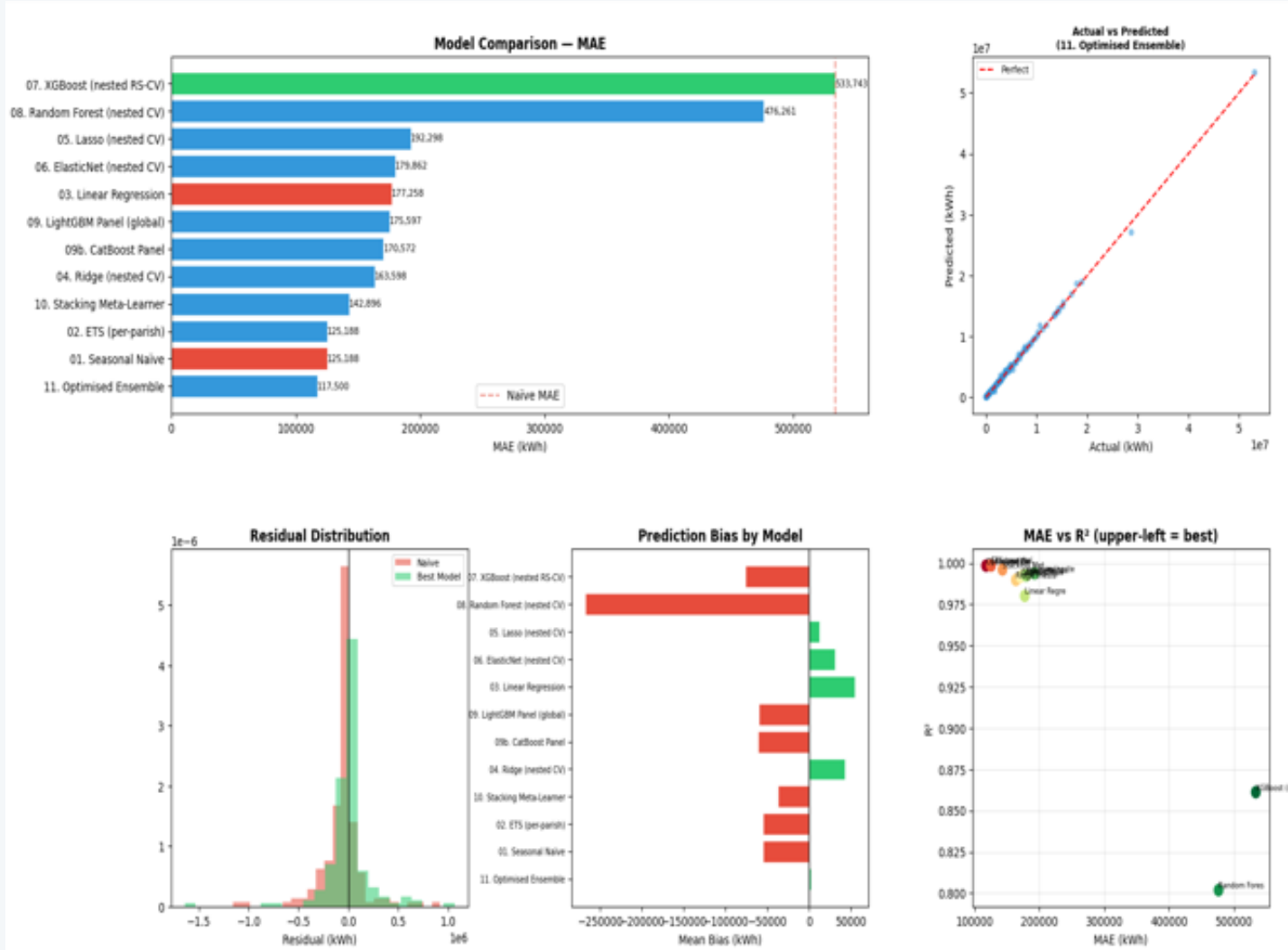
All other 9 candidates (Seasonal Naive, ETS, Ridge, Lasso, ElasticNet, XGBoost, RF, LightGBM Panel, CatBoost Panel) received weight ≈ 0 .

Sensitivity to loss function

- MAE and MAPE elect identical: scale-independence confirmed.
- RMSE-optimal shifts weight to ETS to hedge occasional large LR errors: costs +12% MAE.
- MdAE-optimal spreads to tree models for median robustness: costs +31% MAE.
- MAE choice gives up ~5.5% RMSE vs RMSE-optimal: acceptable trade-off for mean-error planning.



MODEL EVALUATION DASHBOARD



Key Insights

Best ML Model

Optimised Ensemble

MAE: 67,589 kWh R²: 0.998

Strong Benchmark

Seasonal Naïve

MAE: 103,469 kWh R²: 0.998

Beaten by Ensemble (−35 %)

Panel Sensitivity Lost

CatBoost MAE: 130,059 kWh

LightGBM MAE: 203,621 kWh

Both above the ensemble: no panel advantage

Bias & Fit

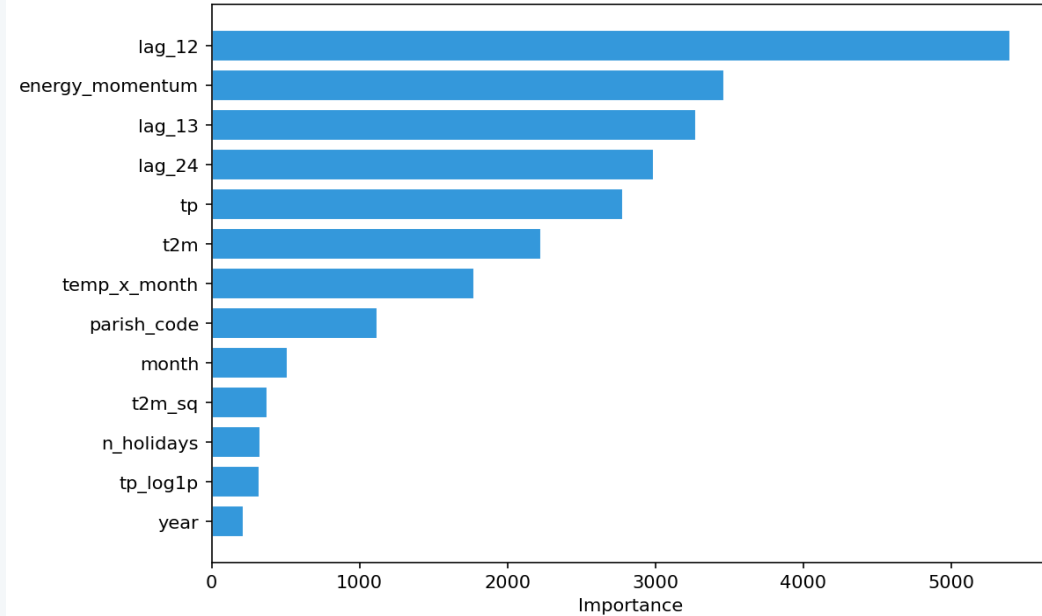
Ensemble: lowest bias, best scatter

XGBoost & RF: >200 kWh underprediction

FEATURE IMPORTANCE ANALYSIS

LightGBM Panel (MAE 204k) vs. XGBoost (MAE 224k): shared signals reveal the true drivers.

LightGBM Panel — Feature Importance (split gain)



#1 lag_12 (LightGBM)

5,300 split gain

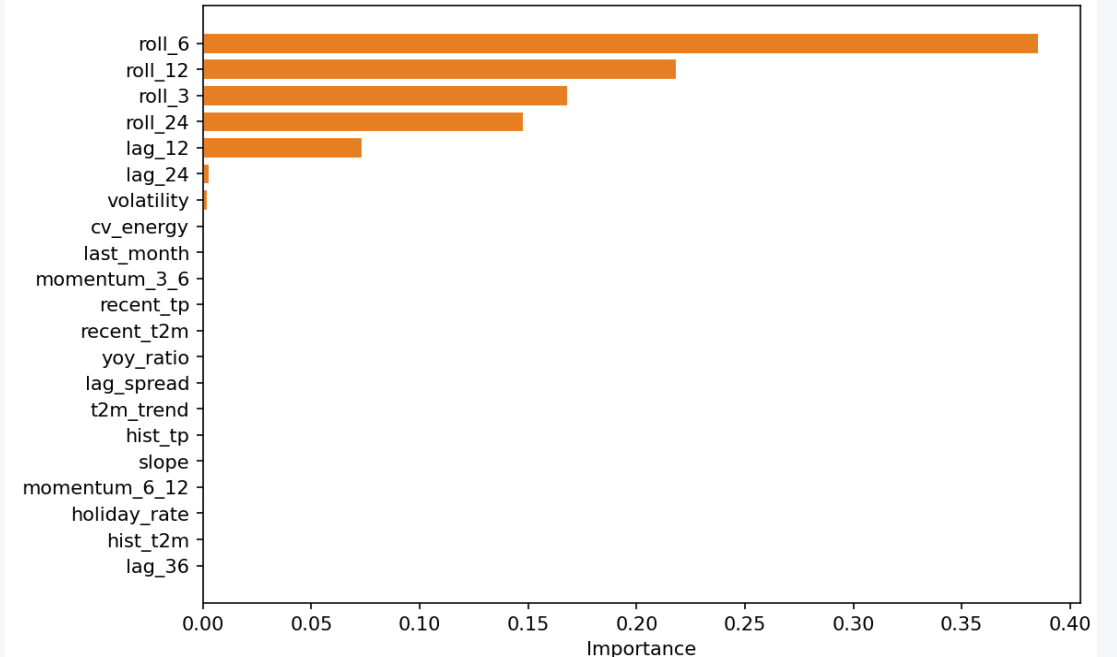
Annual seasonality is the dominant signal

#2 energy_momentum & lag_13/24

Trend momentum + multi-year lags

Parishes hold consumption trajectories 2+ years

XGBoost (Cross-Sectional) — Feature Importance



#1 roll_6 (XGBoost)

6-month rolling avg (0.38 gain), baseline level outweighs seasonal pattern in cross-sectional data

Weather is secondary

tp & t2m in LightGBM top-6

Climate adds signal, but history explains most variance

MODELLING: CONCLUSIONS & TAKEAWAYS

RECOMMENDED COUNTERFACTUAL

Optimised Ensemble (SLSQP blend: Linear Reg 63% + Parish Mean 37%)

MAE 67,589 kWh · MAPE 3.54 % · R^2 0.998 · -25 % MAE vs Parish Mean (90,259), -35 % vs Seasonal Naive (103,469) on the same 87 validated parishes.

A REAL ML MODEL WINS

- Three independent diagnostics confirmed the 2-model blend is correct (SLSQP, greedy selection, L2-regularised re-run all converge).
- Feature importance explains why: lag and level signals (roll_12, lag_12, roll_24) dominate the predictive power, exactly the structure Linear Regression captures cleanly.

STRICT FORWARD CV IS HONEST

- Every model scored on the same held-out 87-parish common subset, with strictly older-CER parishes used for training.
- The ensemble wins on that honest ground, not on a leaky benchmark.

LIMITATIONS

- Validated on 87 of 173 parishes (~50%): the 86 earliest-CER parishes are always in the training window under strict forward CV and never receive an out-of-fold prediction.
- On the common subset Parish Mean (90k MAE) actually beats Seasonal Naive (103k): the strongest naive depends on the eval slice.
- Panel sensitivity check did NOT outperform the cross-sectional setup.

Hand-off to Impact Analysis

These Optimised-Ensemble counterfactual predictions feed directly into the control-vs-treatment framework computed in the next section.

SUPPLEMENTARY: 12-MONTH VECTOR FORECAST (TASK B)

SUPPLEMENTARY

Primary task throughout this report is the annual mean counterfactual (Task A). Task B reframes the same problem as a 12-dimensional multi-output regression: predicting each of the 12 monthly values individually rather than their mean. Same feature matrix, different target structure.

Key Insights

Seasonal Naive dominates at all horizons

Lowest average MAE across all 12 steps, consistent with the main Task A finding that seasonal persistence is the primary signal.

Ridge competitive at steps 1-2 only

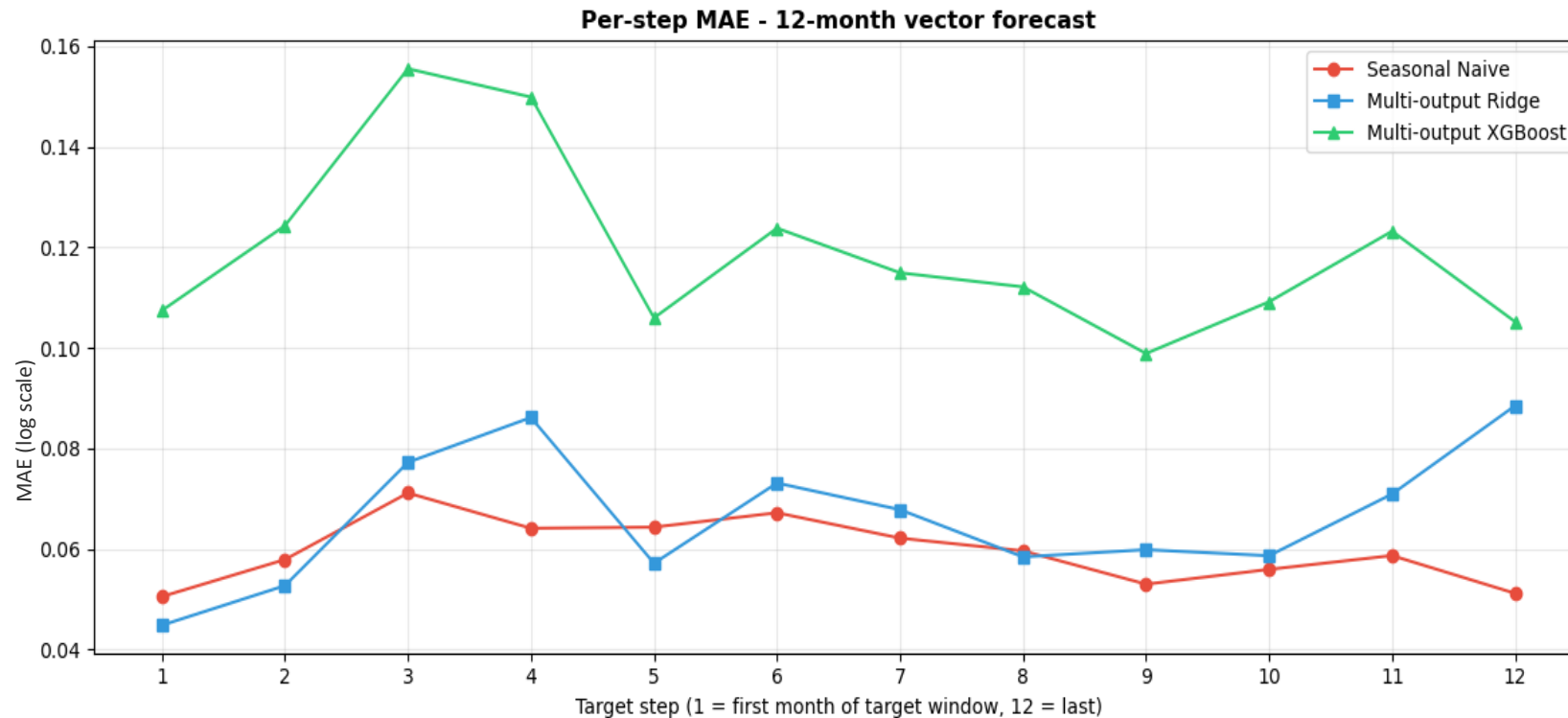
Marginal gains at short horizons (MAE 0.045 vs 0.051) but degrades by step 9, offering no advantage over the full window.

XGBoost fails across all steps

MAE 0.10-0.16 throughout vs Naive's 0.05-0.07: N=173 cross-sectional overfitting is severe on a 12-dimensional target.

Consistent with Task A

Same model ordering holds: Naive > Ridge > XGBoost. Monthly granularity does not change the hierarchy. The annual-mean framing of Task A is the right choice.



Task B confirms the Task A finding: seasonal persistence dominates. Monthly granularity adds complexity without changing the conclusion.

```
graph LR; A[Base Learner Training  
Pre-community data only] --> B[Ensemble Optimization  
SLSQP weight search]; B --> C[Counterfactual Prediction  
Prediction on control parishes]; C --> D[Model Error Estimation  
Observed vs. predicted per parish]; D --> E[Impact Assessment  
Bias-corrected net impact];
```

The flowchart illustrates the process of Modelling and Impact Analysis, consisting of five sequential steps:

- Base Learner Training**
Pre-community data only
- Ensemble Optimization**
SLSQP weight search
- Counterfactual Prediction**
Prediction on control parishes
- Model Error Estimation**
Observed vs. predicted per parish
- Impact Assessment**
Bias-corrected net impact

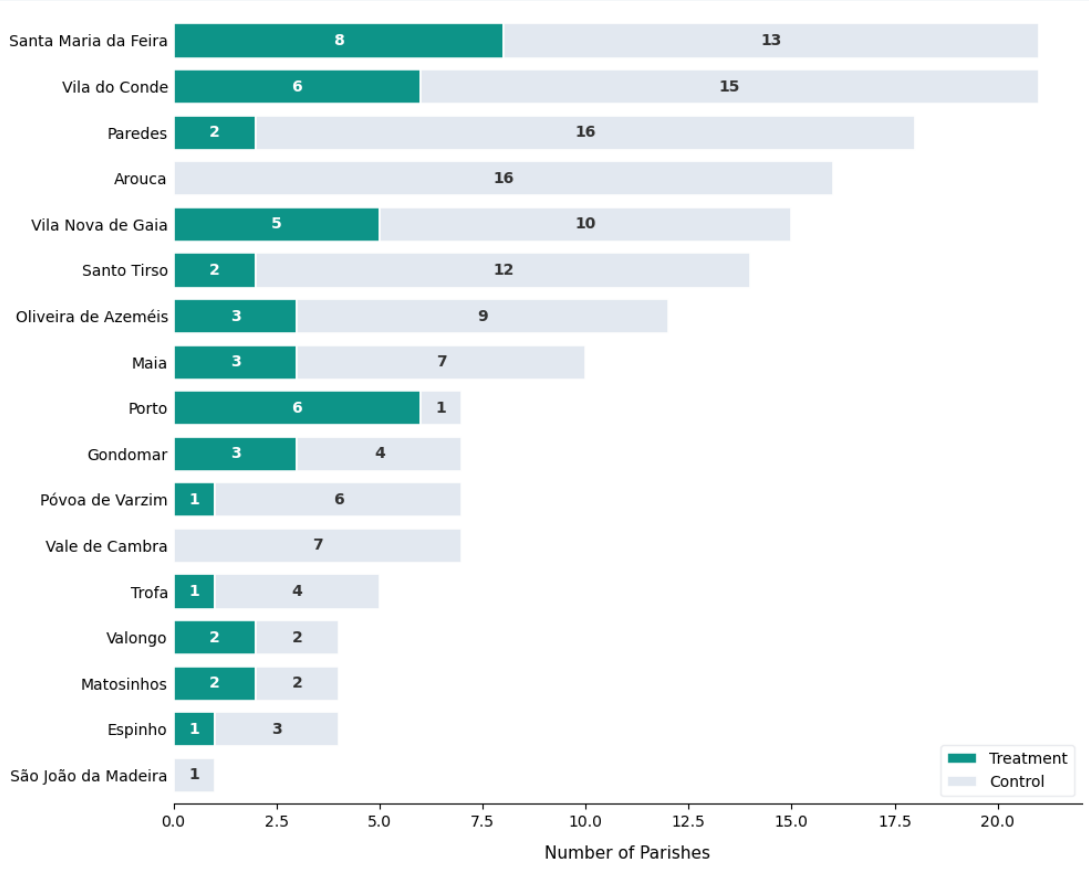
The process is divided into two main phases:

- Modelling** (Steps 1 and 2)
- Impact Analysis** (Steps 3, 4, and 5)

Model Error $error_{pct} = \frac{Y_{real} - \hat{Y}}{\hat{Y}} \times 100$ (For control parishes) How far the model deviates when there is no community	Raw Shift $observed_impact_{pct} = \frac{Y_{real} - \hat{Y}}{\hat{Y}} \times 100$ (For treatment parishes) Observed change including both true impact and model bias	Net Impact (Bias-corrected) $net_impact_{pct} = \overline{error_{pct}} - \overline{observed_impact_{pct}}$ The actual structural effect of energy communities
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IMPACT ANALYSIS: OVERVIEW & METRICS

Control vs. Treatment Parishes



3 municipalities without **treatment** parishes:

- ✓ Arouca
- ✓ Vale de Cambra
- ✓ São João da Madeira

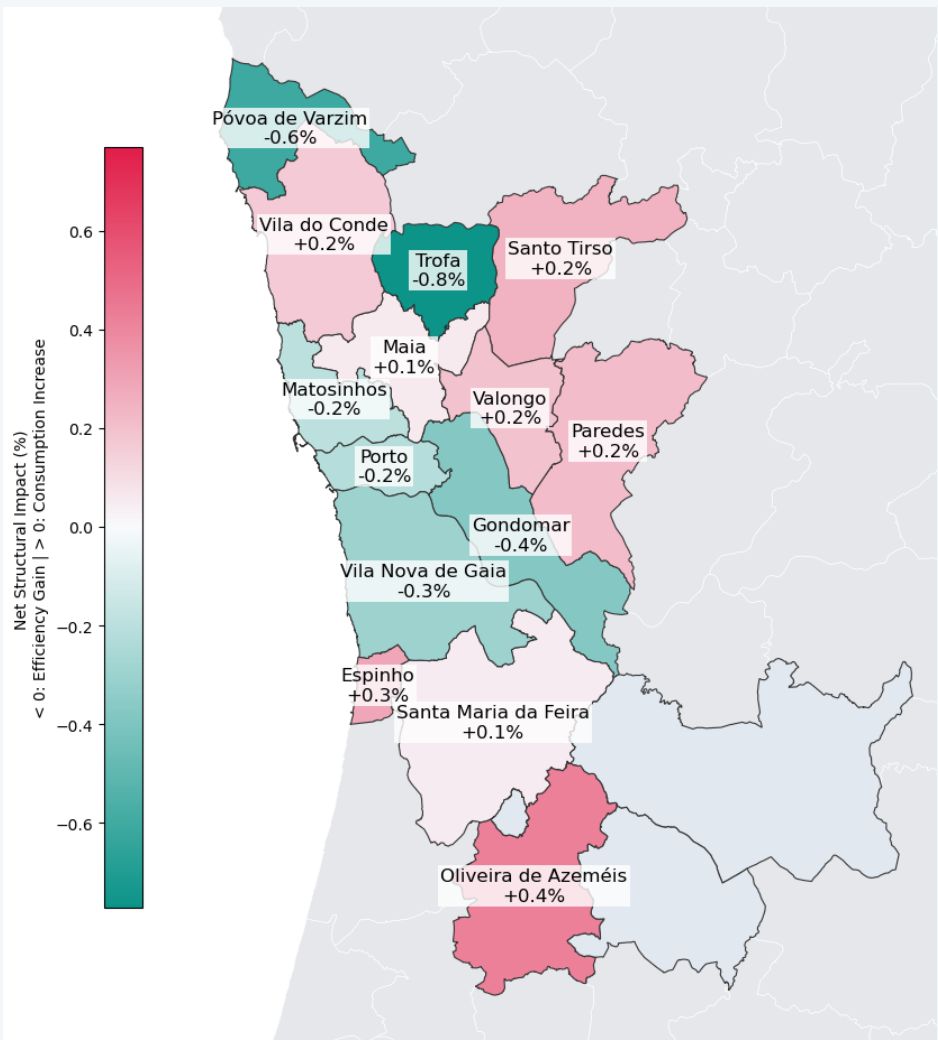
Municipalities with more treatment parishes → ideal for impact analysis

- ✓ Santa Maria da Feira
- ✓ Vila do Conde
- ✓ Porto

IMPACT ANALYSIS: RESULTS

By Municipality

By Parish



High impact variability within the Metro Porto area

6 municipalities with **negative** impact ($< -0.1\%$)
6 municipalities with **positive** impact ($> 0.1\%$)

Impact displays no clear pattern

No discernible pattern based on population density, location or other variables

Impact on energy consumption ranges from:

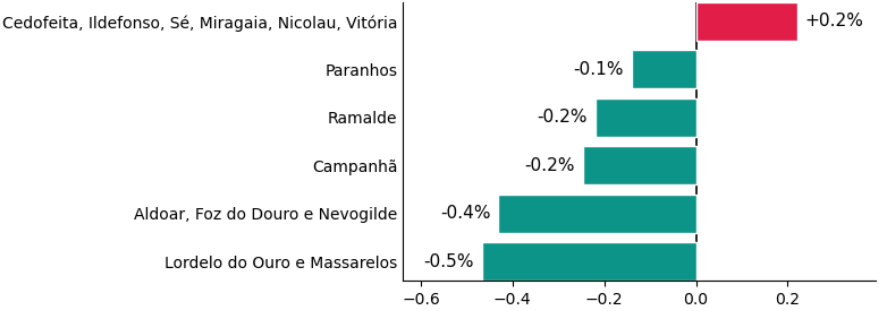
0.8% reduction in Trofa

to 0.4% increase in Oliveira de Azeméis

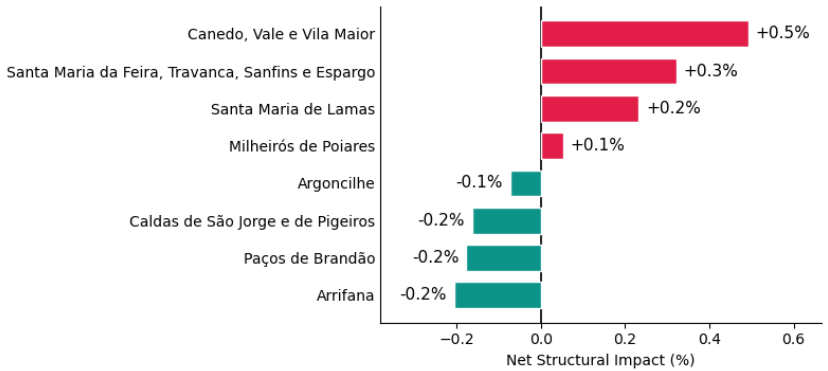
IMPACT ANALYSIS: RESULTS

By Municipality

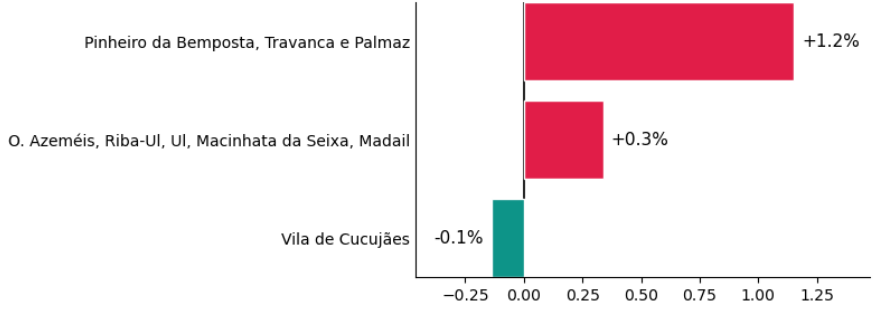
By Parish



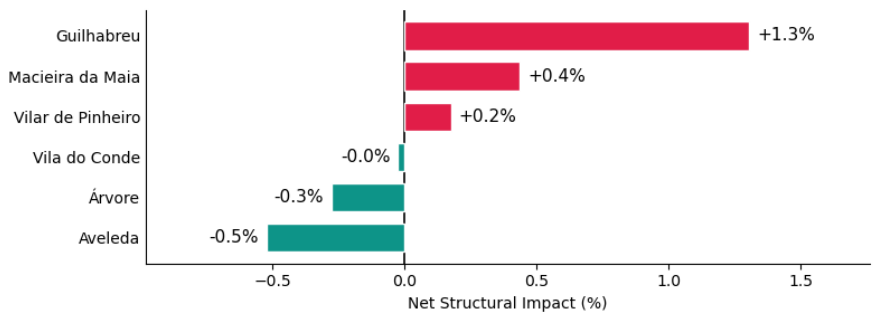
Porto



Santa Maria da Feira



Oliveira de Azeméis



Vila do Conde

Variability is kept at the parish level

All observed municipalities have parishes with opposing impacts

Oliveira de Azeméis

Only 3 treatment parishes, 2 of which have a **positive** impact

Porto

The majority of parishes with energy communities displayed an efficiency gain.

IMPACT ANALYSIS: HYPOTHESIS TESTING

Impact of Energy Communities – Significant or Not?

Normality + Homoskedasticity Diagnostic Tests

Shapiro-Wilk normality test + Levene's homoskedasticity test

Sample Mean Comparison Tests

Mann-Whitney U test (non-parametric 2 sample comparison)

Volume-Weighted Least Squares Regression

2 scenarios: all data (scenario 1) and outlier filtering (scenario 2)

Actual vs. predicted comparison

Parish **weights** based on **consumption** levels

Final Statistical Verdict

Not Significant

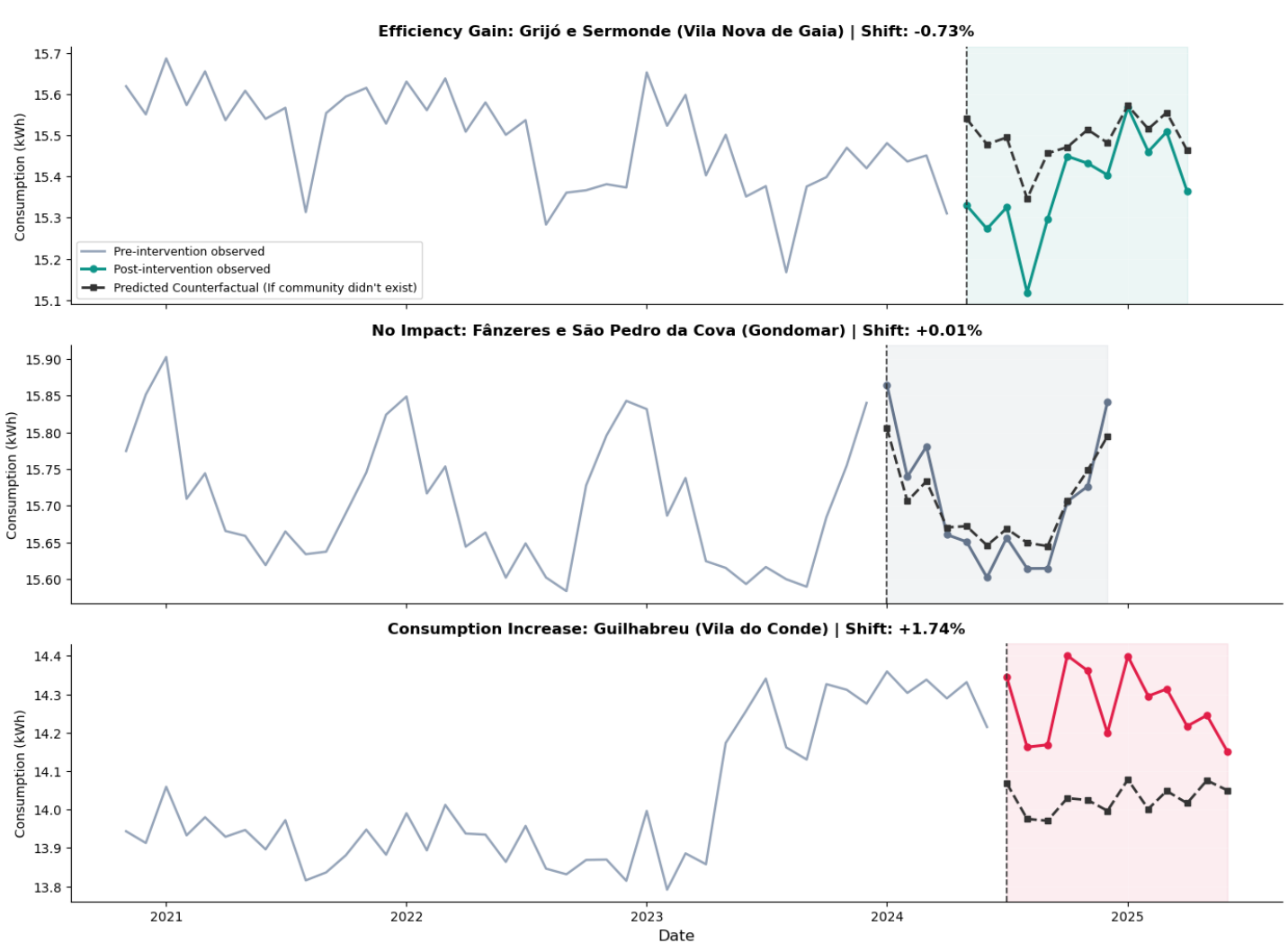
p-value = 0.303

Not Significant

Scenario 1: p-value = 0.386

Scenario 2: p-value = 0.153

CER IMPACT: DISCUSSION



Consumption Decrease

Observed < Predicted

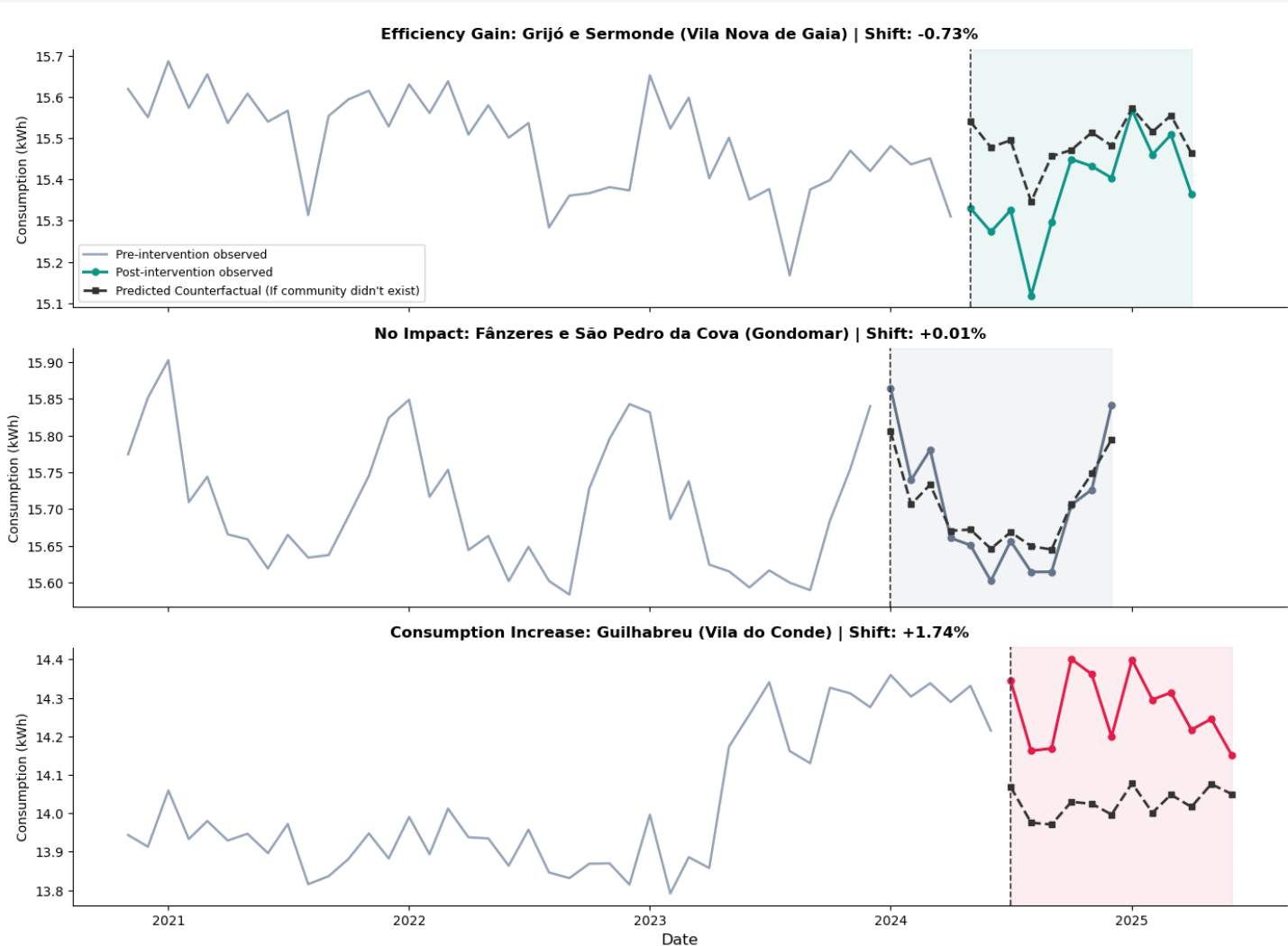
Consumption Maintained

Observed = Predicted

Consumption Increase

Observed > Predicted

CER IMPACT: DISCUSSION



1

Consumption “noise” surpasses CER impact

2

Energy consumption is a complex feature

It displays seasonality

Depends on:

- Environmental conditions
- Population size and behaviour
- Economical context

3

12-month time frame is not enough

Energy communities are expected to have a **long-term** effect

Not captured by first 12 months

CONCLUSIONS

DATASET & CONTEXT

- **The E-redes open data portal** fosters analytical challenges that drive improvements in data quality, safety awareness, and evidence-based decision-making.
- **Energy communities** offer benefits such as reduced **energy costs**, increased local **renewable** generation and greater **energy independence** for members. Their full effects require longitudinal study over a longer time horizon to be reliably detected.

IMPACT ANALYSIS

- **Impact** results were highly mixed and statistically **non-significant**, likely because:
 - **observation window** covers only **12 months** post-creation (insufficient to capture the long-term behavioural shifts).
- **Energy consumption** is inherently complex and highly variable:
 - short-term fluctuations (e.g. driven by seasonal and economic factors) may mask the effect of **energy communities**.
- Granular **power quality** data at parish level would have enriched the impact analysis and improved interpretation of the results.

MODELLING

- An optimized **ensemble** of **linear regression** and **parish mean** proved to be the strongest predictors of energy consumption, reflecting the data's cross-sectional structure, where level and lag signals dominate over complex non-linear patterns.
- Reliable **energy consumption forecasting** demands careful handling of temporal **data leakage**. A custom chronological forward cross-validation (ChronoForward CV) was developed to ensure strictly honest out-of-sample evaluation.

FURTHER WORK

- **Long-term** impact studies (5- and 10-year horizons) to capture the sustained effect of energy communities on consumption behaviour.
- **Geographical segmentation** studies (e.g. rural vs. urban): infrastructure, industry mix, and consumption profiles differ substantially, and energy communities may have very distinct impacts across these contexts.
- Incorporation of **socioeconomic indicators** (income, employment, housing type) to better explain heterogeneity in adoption and impact across parishes.

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